

* Always Remember: → Try Your Best to come up with the Recursive logic only, Because after that DP is just 5 minutes of work. — vHixem

* Problem: —

→ could also say stick of length n

Given an integer n denoting the Length of a line segment. You need to cut the line segment in such a way that the cut length of a line segment each time is either x , y or z . Here x , y , and z are integers.

After performing all the cut operations, your total number of cut segments must be maximum. Return the maximum number of cut segments possible.

Note: if no segment can be cut then return 0.

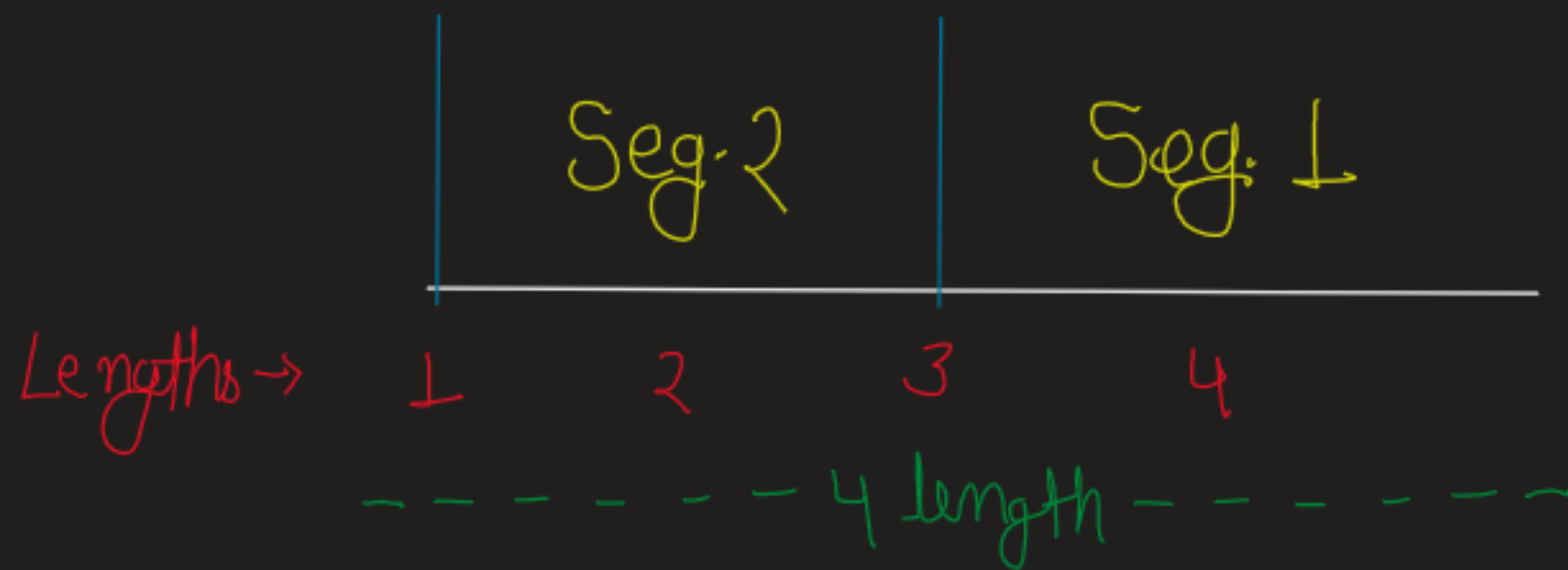
Input: $n = 4, x = 2, y = 1, z = 1$

Output: 4

$n = 4$

— — — — 4 length — — — —

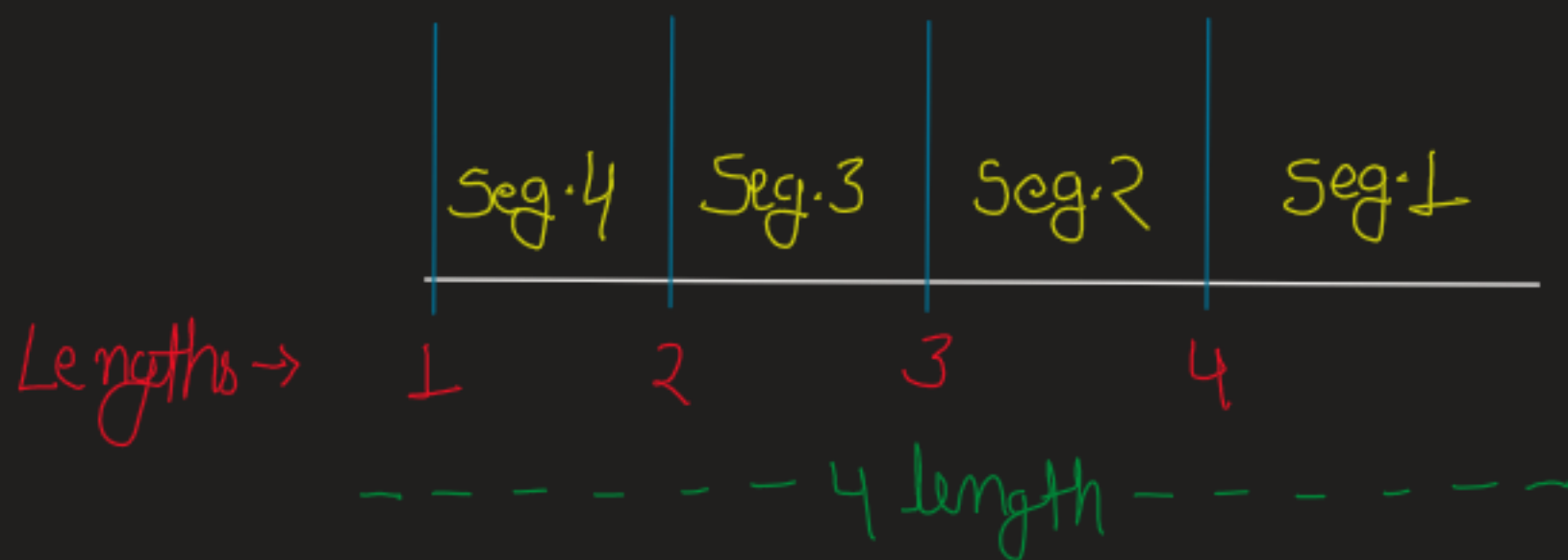
① → suppose we cut in length $x=2$
→ 2 segments possible



start cutting from behind
or front, overall answer
will be the same.

(means go n to 0)

② → suppose we cut in length $y=1$
→ 4 segments possible



or
(0 to n)

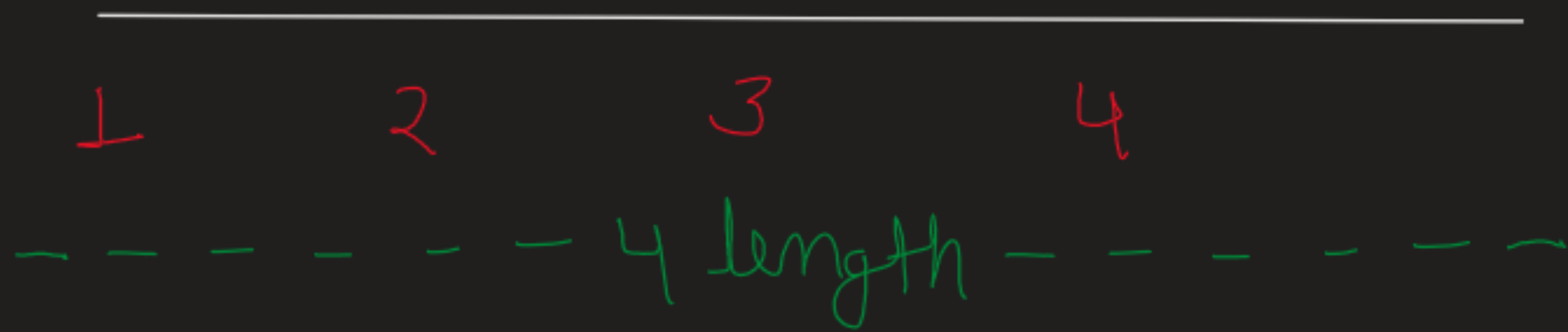
↓
leads to same
result

→ from both ways maximum segments possible
is $\max(2 \text{ segments}, 4 \text{ segments})$

hence answer is ④

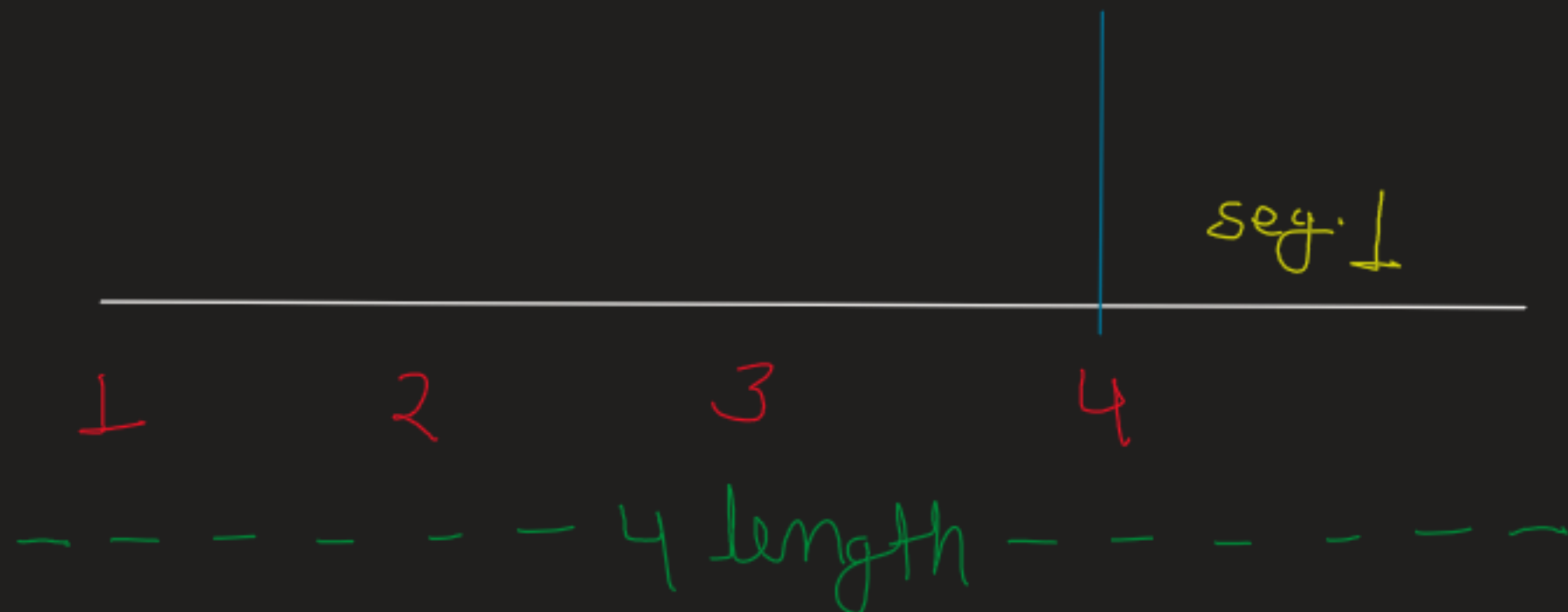
~ Let $n=4 \rightarrow$ Max 4 segments possible

$\rightarrow$ suppose we cut in length $y=1$

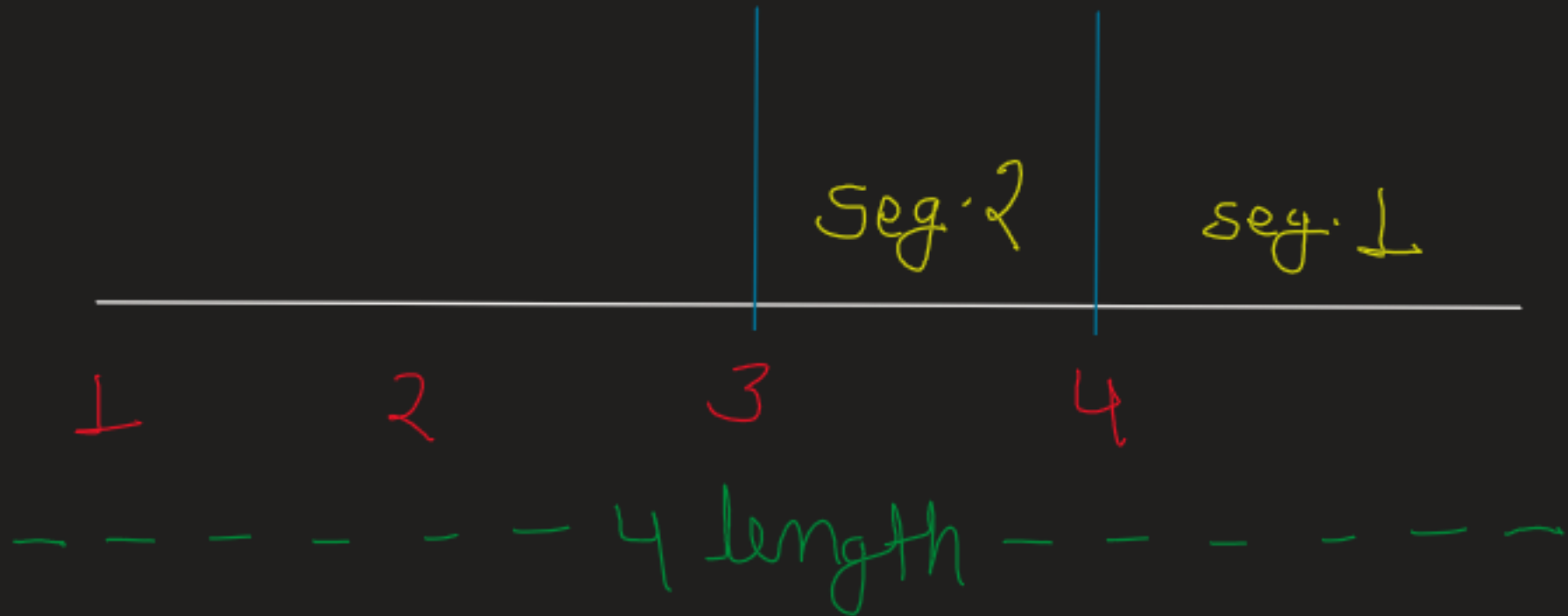


(lets start cutting from behind)

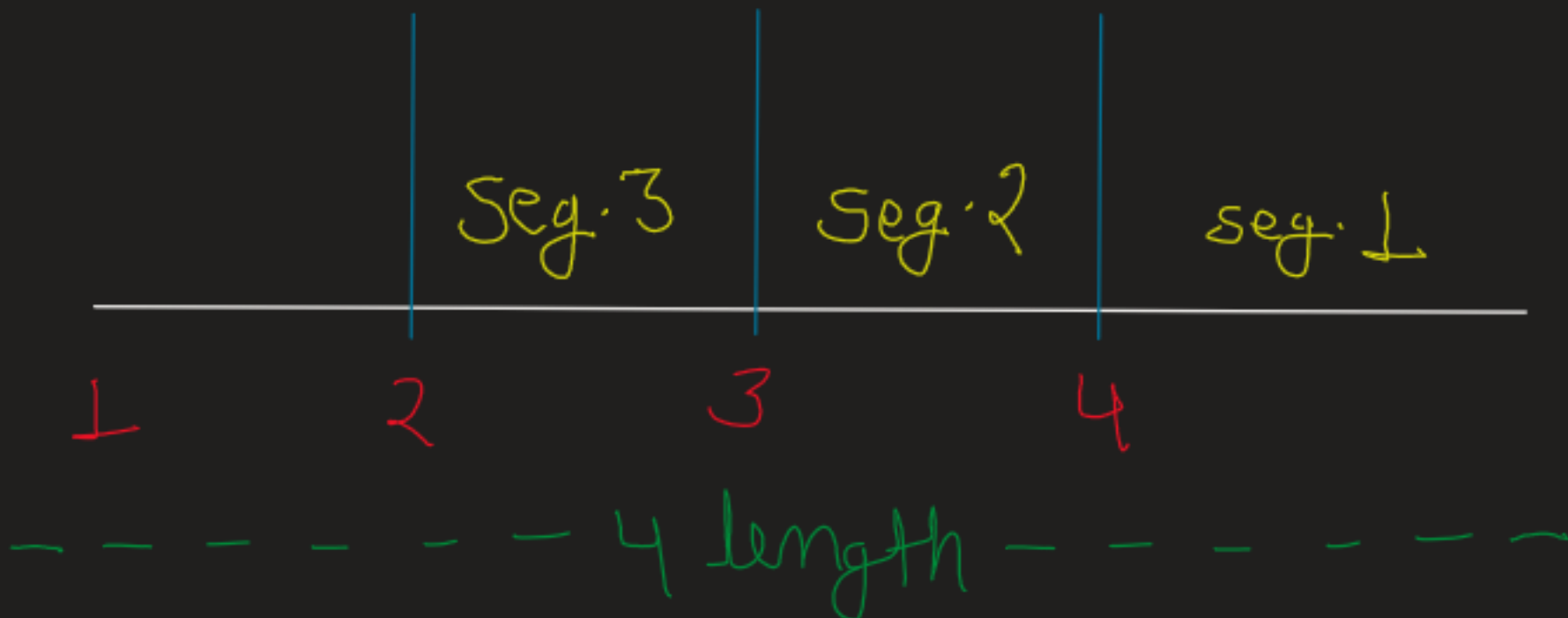
$\rightarrow$ now $n=3$ remains:- (cut in $y=1$)



⇒ now $n=2$ remains :- (cut in $y=1$)



⇒ now $n=1$ remains :- (cut in $y=1$)

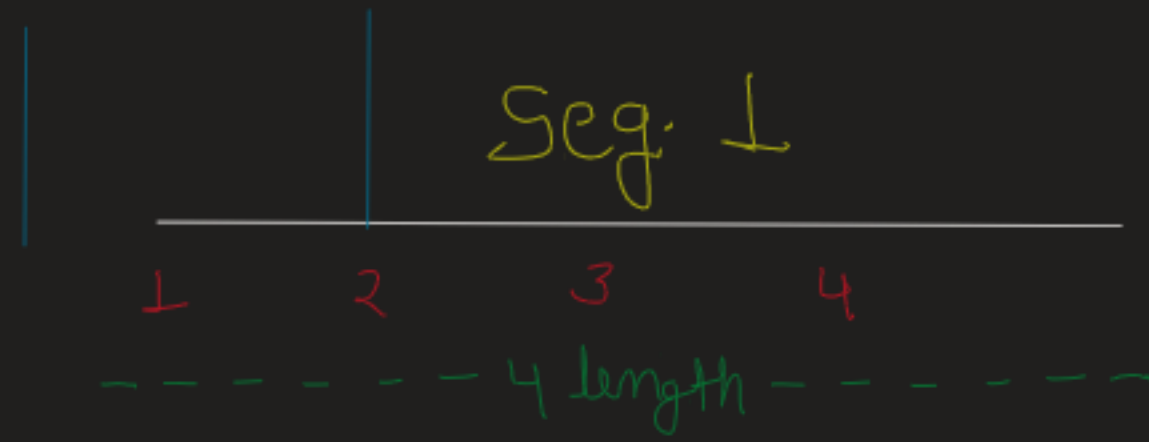


→ now $n=0$ remains:- (cut in $y=1$)



→ here we made 4 segments

→ now n is negative :- (cut in $y=2$)



→ This way of cutting is not valid
because (n) is becoming negative (logically no stick exists before $+$)

here, we did this:-

$$n = n - x$$

$$n = n - y$$

$$n < 0 \text{ now}$$

$$n = 4 - 3 \Rightarrow 1$$

$$n = 1 - 2 \Rightarrow -1$$

$$1 - 1 = 0$$

→ So when a way of cutting becomes invalid
no maximum segments possible because stick is
not properly cut.

